# 1. Prime-field separator-rank lead reconciliation

2026-11-01 · search and implementation report · no scientific run

## 1.1. Outcome

The requested next discriminator was separator rank or small-root growth as relation arity increases, with special-prime structure as a second lead. The repository already preregisters the more faithful separator experiment: contract the actual hashed Boolean network at a semantic cut and compute exact balanced flattening ranks against topology-preserving rewired nulls.

This round implemented the missing exact census core. It did not generate an E1 row. The protocol stays draft until production cuts, matched nulls, fresh-process A/A, power, resource limits and image identities are frozen.

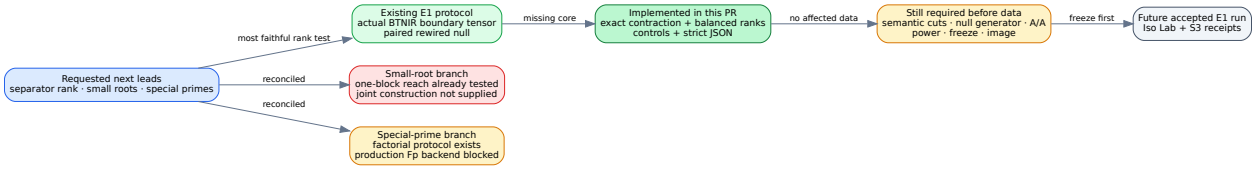



Figure 1: Requested leads, existing evidence, and the implemented slice. Green is verified development implementation; amber is an open gate; red is a closed or unsupplied route at the stated scope.

## 1.2. Evidence reconciliation

| Record | Decision-relevant evidence |
|---|---|
| Prime-field breakthrough map | The convex one-block Semaev lattice is already measured below the exponent required by the complete relation/linear-algebra bridge. |
| Digit presentation | Last fall degree grows; at toy scale the Macaulay space spans 60–100% of the ring and a curve-free block null reproduces the early fall. |
| Tensor–Boolean E1 | Exact boundary ranks against rewired nulls were already the registered next experiment; `tensor-census` was the named missing implementation. |
| 31 October lead review | Ranks the hashed tensor route first and defines a separate factorial special-prime/special-curve control. |
| Special-prime constructions | 22 certified constructions found no small embedding degree; this does not answer Boolean-representation compression. |

The earlier special-prime scratch comparison was not imported: curves and group orders differed, there were only two curve draws per prime type, and six searched predicates had no frozen family-wise rule. The earlier group-sum support proxy was also not imported: it does not contract the actual BTNIR object or use a wiring-matched null.

## 1.3. Implemented exact instrument

The native core:

- validates canonical IR and a nonempty proper vertex side;
- leaves canonical cut edges open and rejects cut pins;
- enumerates all admitted free internal assignments under an explicit cap;
- evaluates the digest-bound truth table or typed program at every selected predicate;
- emits exact arbitrary-precision boundary counts, support and a stable tensor digest;
- computes canonical and exhaustive balanced ranks with fraction-free integer elimination; and
- records big-integer operation counts and strict JSON provenance.

Development checks pass for ranks 1, 2 and 4; a canonical equality network; cap and pin failures; and 1,152 deterministic matrices cross-checked against rational Gaussian elimination. Rust 1.78 focused release tests and CLI build pass. Current-stable all-target Clippy passes with warnings denied.

## 1.4. Request status and next gate

| Item | Status | Evidence or gap |
|---|---|---|
| Separator-rank census | Instrument complete | Production cut/null generator and frozen ladder remain open. |
| Small-root growth | Existing scoped evidence | No new joint-system construction was supplied. |
| Special-prime test | Backend blocked | Factorial protocol exists; production Fp theory backend does not. |
| Subexponential claim | Not established | Finite rank data cannot prove an asymptotic family bound. |

Next: add stable semantic-region metadata, generate the primary cut after slot `ceil(m/2)`, implement constrained double-edge-swap nulls and fresh-process A/A, then freeze the final ladder and run it on Iso Lab from a committed digest-pinned image.

The canonical scoreboard, progress timeline, leaderboard and curve browser remain unchanged because this round adds no admitted numeric comparison, curve, factor base, phase cost or reference change.

## 1.5. Sources

Petit–Kosters–Messeng, PKC 2016; Bochum `subexponential-ecdl` v0.2 at the revision pinned by the lead review; OpenAI Appendix D at its pinned revision; `RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md`; and the repository studies named above. Exact paths and requirement mapping are in `SEARCH_20261101.md` and `E1_IMPLEMENTATION.md`.